SAM SEIFI

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EDUCATION

BSc Physics with Astrophysics and Cosmology, King's College London — Sep 2025 – expected 2028

- **Predicted:** First Class.

GCE A-levels, The Compton Sixth Form — Sep 2023 – Aug 2025

- **Grades:** Mathematics (A), Physics (B), Computer Science (B).

RESEARCH EXPERIENCE

The Hubble Constant Across Four Cosmological Probes

- Developed **four independent CosmoSIS inference pipelines** to investigate the Hubble tension using **DES-Y3 BAO, DESI DR2 BAO, Pantheon+SH0ES and compressed Planck 2018 CMB** constraints, comparing their sensitivity to cosmological assumptions and dataset coverage.
- Implemented **curved wCDM models** with continuous sampling of H_0, Ω_m, Ω_k and w. Integrated **CAMB sound-horizon calculations**, Bayesian MCMC inference and physically motivated parameter bounds, including cosmological density-closure validation.
- Analysed marginalised posterior distributions, parameter degeneracies and differences between BAO, supernova and CMB constraints. Developed **autocorrelation-based convergence diagnostics**, posterior visualisations and comparative statistical analyses, identifying limitations of unconverged chains and approximate likelihood models.

SELECTED TECHNICAL PROJECTS

Hubble Tension – Supernova Cosmological Inference

- Constructed a **flat ΛCDM parameter-estimation pipeline** using **1,657 Pantheon+SH0ES supernova and Cepheid-calibrator records**, incorporating statistical and systematic covariance. Compared **emcee MCMC and BFGS maximum-likelihood optimisation** to estimate the calibrated expansion rate and evaluate statistical uncertainties and method limitations.

EvoXRB: X-ray Binary Spectral-Timing Inference

- Developed a reproducible X-ray spectral-timing pipeline combining **genetic optimisation**, local refinement, posterior inference and parameter recovery. Achieved **100% convergence across 60 synthetic fits** and **98.95% mean parameter-recovery accuracy** across 240 estimates.

Exoplanet Orbital and Thermal Modelling

- Processed **149,648 orbital-quarter records** into **6,284 exoplanet–host systems**. Implemented a **361-sample Keplerian simulation** per orbit, calculating orbital distance, vis-viva velocity and equilibrium-temperature variation.

Hertzsprung–Russell Diagram from Gaia DR3

- Queried and processed **one million stellar sources** from ESA Gaia DR3 using Python, ADQL and Astroquery to construct and visualise an H–R diagram.

Earthaccess: Open-Source Contributor

- Contributed documentation to the **earthaccess Python library**, supporting access to NASA Earth science datasets.

Minesweeper AI: Constraint-Based Solver

- Implemented a Python solver using **linear systems, constraint enumeration and probabilistic inference** to identify safe moves, mine locations and unavoidable guesses.

Additional software: Rust/WebAssembly ASCII Art Generator; C# Terraria multiplayer mod.

EXPERIENCE

The Deli Downstairs — Jul 2026 – Nov 2026
Kitchen Porter, part-time — *London, UK*

- Supported high-volume kitchen operations, maintaining hygienic and organised preparation and equipment areas.

National Student Space Conference (NSSC) — Feb 2026
Volunteer — *London, UK*

- Represented the university's space society and assisted with conference events and specialist scientific talks.

MedMart Borehamwood — Jul 2023 – Sep 2024
Pharmacist Assistant, part-time — *London, UK*

- Assisted with medication handling and organisation for elderly patients, coordinating accurate, time-sensitive work with colleagues.

TECHNICAL SKILLS

Scientific computing: Python, NumPy, SciPy, pandas, Matplotlib, CosmoSIS, emcee, Astroquery, Jupyter, Bayesian inference, MCMC, numerical optimisation.
Programming and tools: Rust, C#, JavaScript, HTML/CSS, SQL, Linux, Git, APIs, WebAssembly, LaTeX, Blender.
Laboratory: Oscilloscope, function generator, digital multimeter, experimental data analysis and uncertainty propagation.